

1st EXP obj 1 code and output

```
clc;

clear;

close all;

% Parameters

fs = 1000;      % Sampling frequency
t = 0:1/fs:1;   % Time vector (1 second)
TBW = 600;      % Total Bandwidth (kHz)
Users = 3;      % Number of users
CBW = TBW / Users; % Bandwidth per user

% Carrier frequencies
fc = [100 300 500];

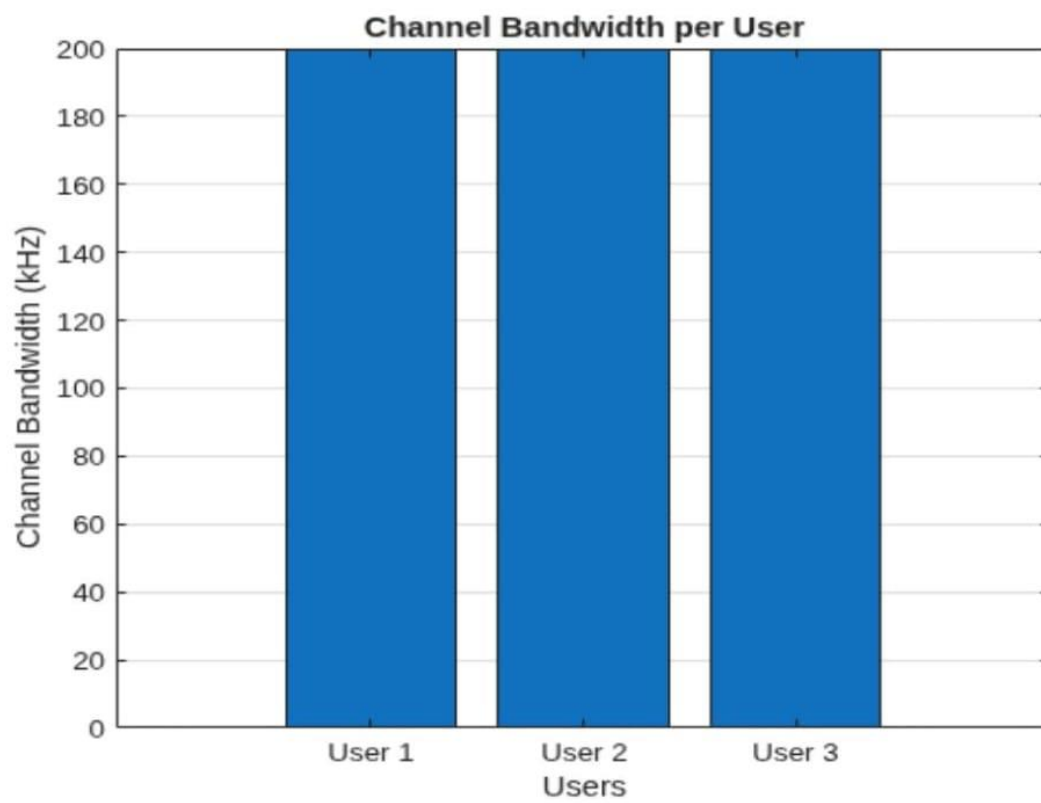
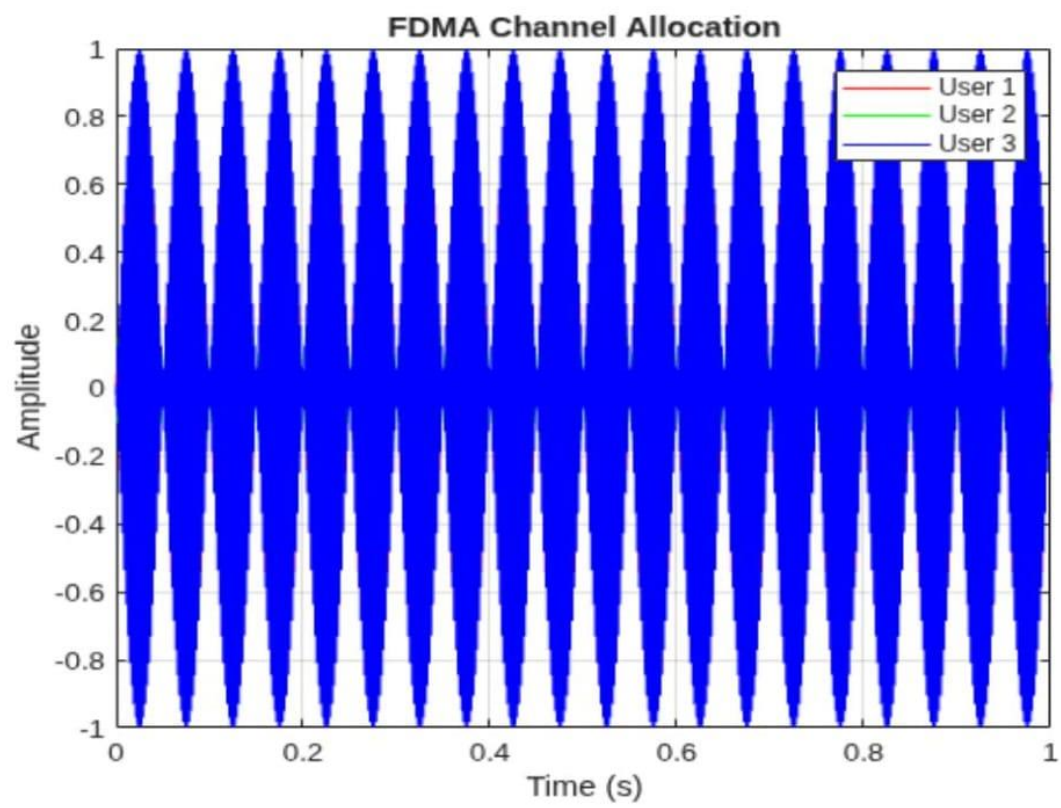
% Message signal
m = sin(2*pi*10*t);

% FDMA signals
s1 = m .* cos(2*pi*fc(1)*t);
s2 = m .* cos(2*pi*fc(2)*t);
s3 = m .* cos(2*pi*fc(3)*t);

% Plot transmitted signals
subplot(2,1,1);
plot(t, s1, 'r', t, s2, 'g', t, s3, 'b');
grid on;
title('FDMA Channel Allocation');
xlabel('Time (s)');
ylabel('Amplitude');
legend('User 1','User 2','User 3');

% Plot channel bandwidth allocation
subplot(2,1,2);
bar(1:Users, CBW*ones(1,Users));
grid on;
```

```
title('Channel Bandwidth per User');  
xlabel('Users');  
ylabel('Channel Bandwidth (kHz)');  
xticklabels({'User 1','User 2','User 3'});
```



EXP 1 OBJ 2 CODE AND OUTPUT

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
```

```
disp(' User Data Rate (kbps)')
```

```
disp([(1:Users)' Rate'])
```

```
figure
```

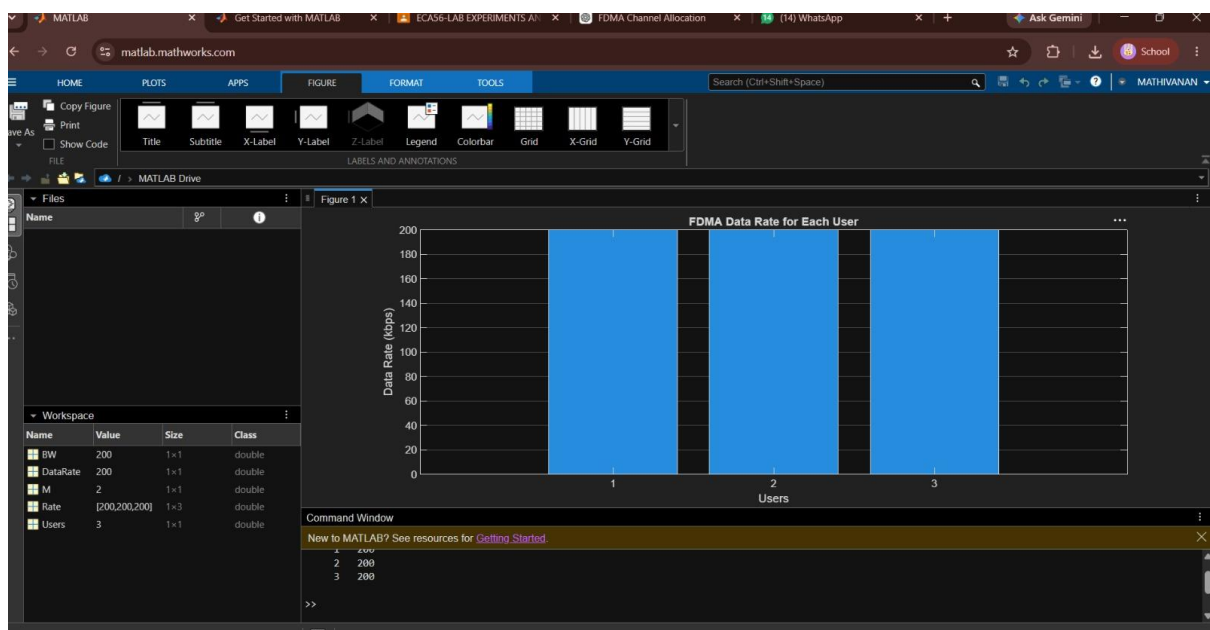
```
bar(1:Users,Rate)
```

```
xlabel('Users')
```

```
ylabel('Data Rate (kbps)')
```

```
title('FDMA Data Rate for Each User')
```

```
grid on
```



EXP 1 OBJ 3 CODE AND OUTPUT

```
clc; clear; close all;
```

```
FrameTime = 4.615; Slots = 8;
```

```
SlotDuration = FrameTime/Slots;
```

```
t = 0:SlotDuration:FrameTime;
```

```
figure
```

```
subplot(2,1,1)
```

```
stem(t,ones(size(t)),'filled')
```

```
title('TDMA Time Slot Structure')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Level')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(t,ones(size(t)),'-o','LineWidth',1.5)
```

```
title('Continuous View of Slot Allocation')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Presence')
```

```
grid on
```

